ESSENTIAL FATTY ACIDS

Dr. Muhammad Ahsan

FATTY ACIDS

Fats (fatty acids) fall into two main groups – saturated and unsaturated – based on their chemistry.

There are three major classes of unsaturated fatty acids: omega-3, omega-6 and omega-9. The omega-6s and omega-3s are essential.

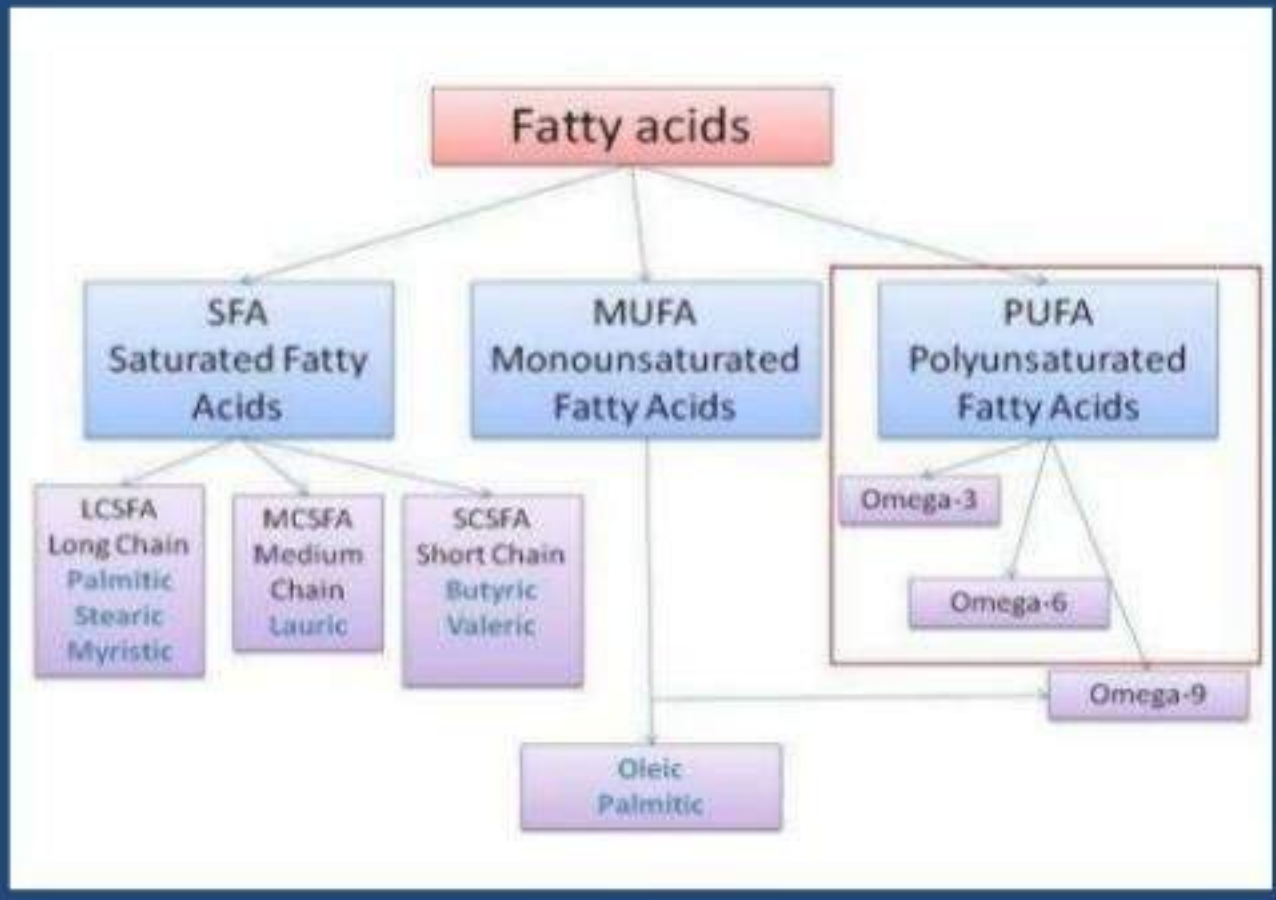
The omega-9s are non-essential because the body can make them from other fatty acids. Unsaturated fatty acids are further classified as either monounsaturated or polyunsaturated.

EFAs are polyunsaturated. They include:

2- The omega-3 fatty acid alpha-linolenic acid (ALA) and its derivatives, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA).

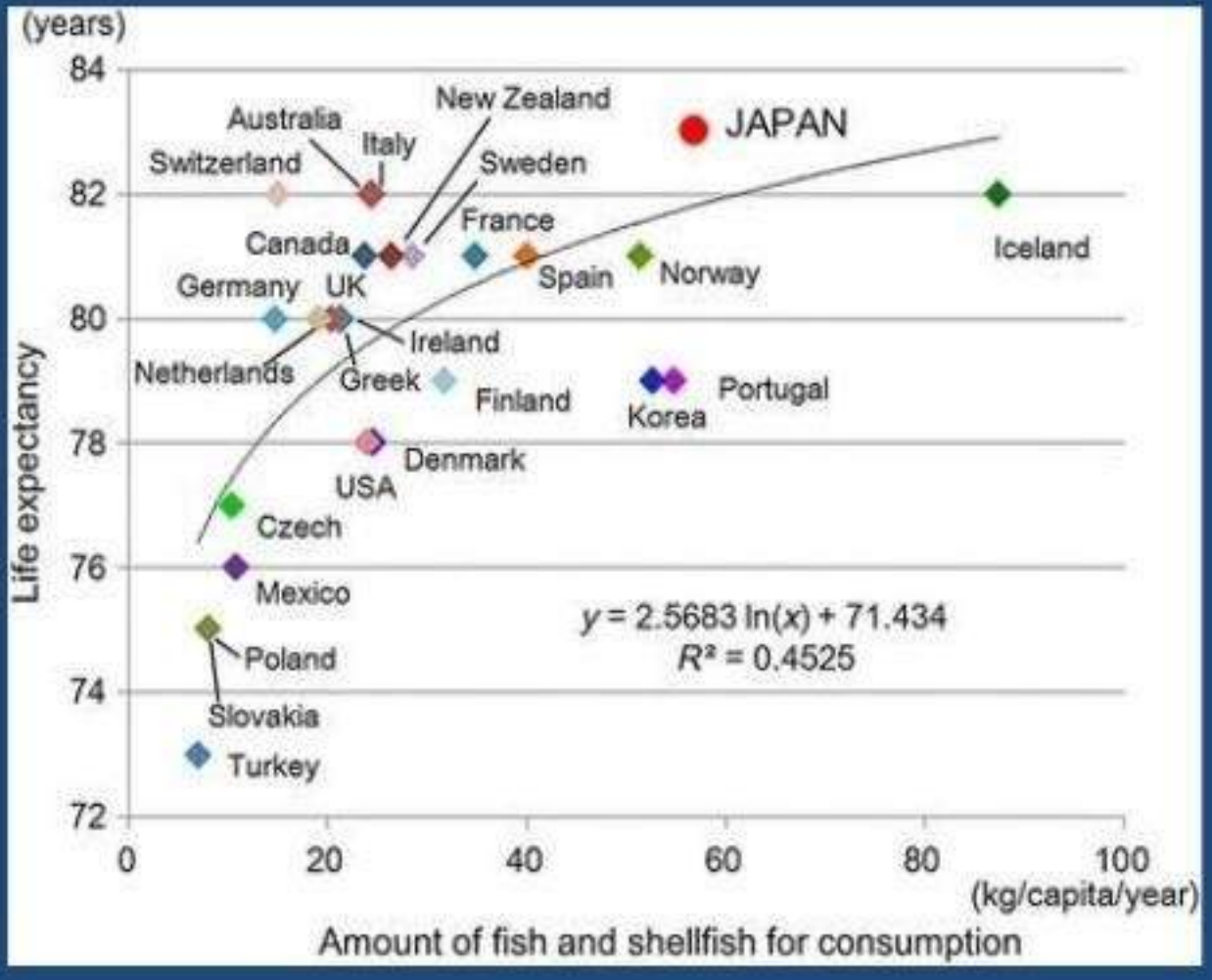
1- The omega-6 fatty acid linoleic acid (LA), and its derivatives, gamma-linolenic acid (GLA) and arachidonic acid (AA)

Theoretically, only LA and ALA are absolutely essential. However, the fatty acids derived from them are also generally considered essential.

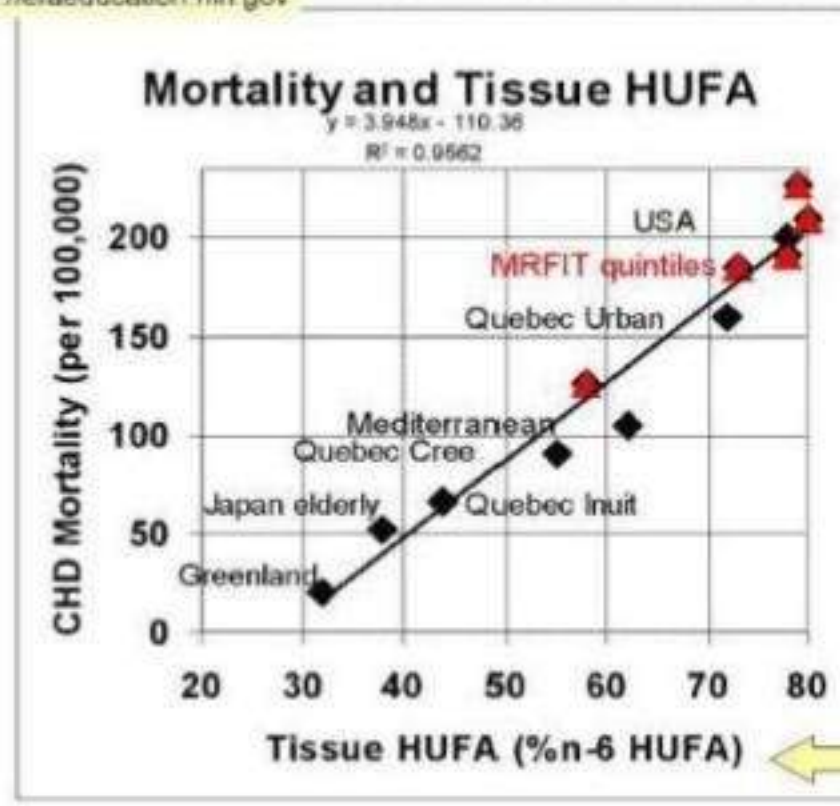

Fatty acids
SFA
Saturated Fatty
Acids
MUFA
Monounsaturated
Fatty Acids
PUFA
Polyunsaturated
Fatty Acids
LCSFA
Long Chain
Palmitic
Stearic
Myristic
MCSFA
Medium
Chain
Lauric
SCSFA
Short Chain
Butyric
Valeric
Omega-3
Omega-6
Omega-9
Oleic
Palmitic

Essential Fatty Acids

- The body can synthesize most of the fats it needs from the diet. However, two essential fatty acids, linolenic and linoleic acid, cannot be synthesized in the body and must be obtained from food.


(years)
84
82
80
78
76
74
72
Life expectancy
0
20
40
60
80
100
(kg/capita/year)
Amount of fish and shellfish for consumption
JAPAN
Australia
Italy
New Zealand
Switzerland
Sweden
France
Canada
Iceland
Spain
Norway
Germany
UK
Ireland
Netherlands
Greek
Finland
Portugal
Korea
Denmark
USA
Czech
Mexico
Poland
Slovakia
Turkey
y = 2.5683 ln(x) + 71.434
R² = 0.4525

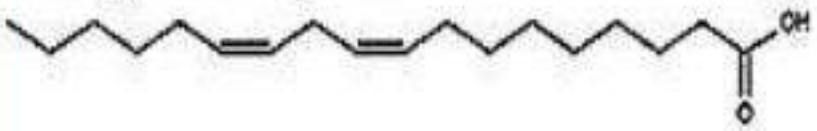
http://efaeducation.nih.gov



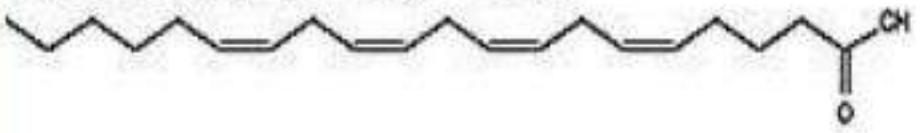
What Mortality Rate is Optimal??

FIGURE 2. Coronary heart disease mortality is proportional to n-6 HUFA in plasma HUFA. Available at <http://efaeducation.nih.gov/sig/personal.html> .

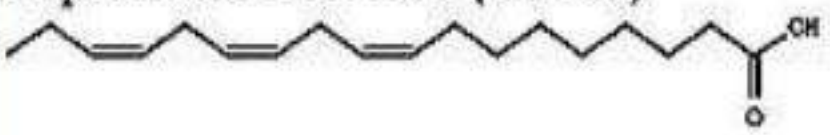
Omega-6 fatty acids: Linoleic acid (C 18:2)



Arachidonic acid (C 20:4)

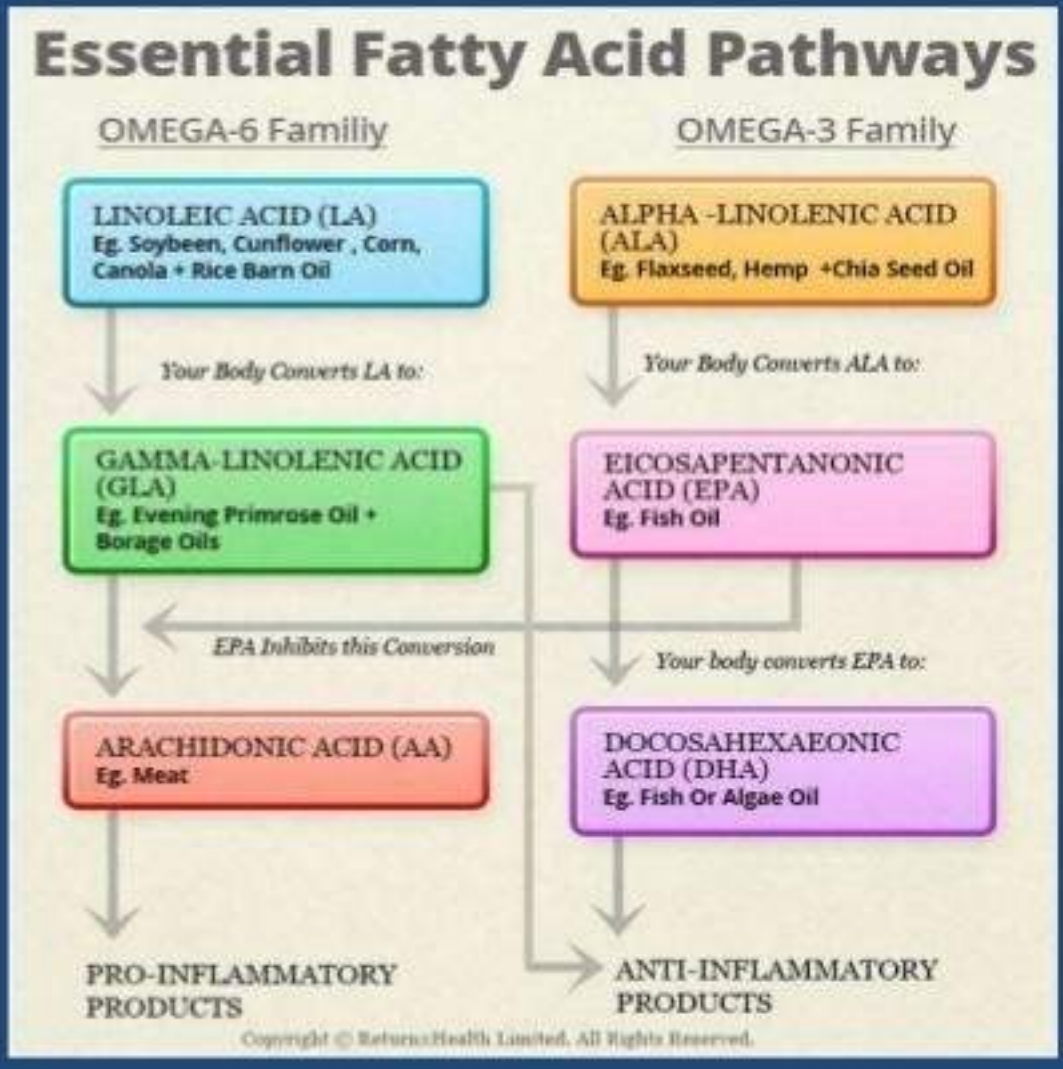


Alpha-linolenic acid (C 18:3)



Docosahexaenoic acid (DHA; C 22:6)




Essential Fatty Acid Pathways
OMEGA-6 Familiy
OMEGA-3 Family
LINOLEIC ACID (LA)
Eg. Soybeen, Cunflower , Corn, Canola + Rice Barn Oil
ALPHA -LINOLENIC ACID (ALA)
Eg. Flaxseed, Hemp +Chia Seed Oil
Your Body Converts LA to:
Your Body Converts ALA to:
GAMMA-LINOLENIC ACID (GLA)
Eg. Evening Primrose Oil + Borage Oils
EICOSAPENTANONIC ACID (EPA)
Eg. Fish Oil
EPA Inhibits this Conversion
Your body converts EPA to:
ARACHIDONIC ACID (AA)
Eg. Meat
DOCOSAHEXAEONIC ACID (DHA)
Eg. Fish Or Algae Oil
PRO-INFLAMMATORY PRODUCTS
ANTI-INFLAMMATORY PRODUCTS
Copyright © Returns Health Limited. All Rights Reserved.

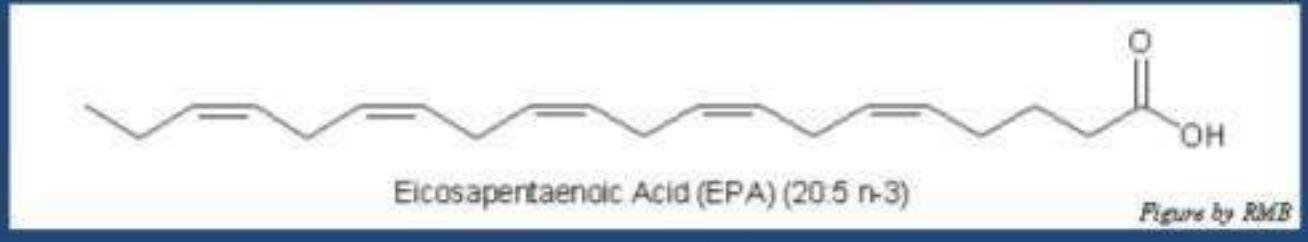
Omega-3 fatty acids

- Omega-3 fatty acids are a type of polyunsaturated fat (like omega-6), considered an essential fatty acid because it cannot be manufactured by the body. As a result, people must obtain omega-3 fatty acids from foods such as fish, nut, and plant-based oils such as canola oil and sunflower oils.

Types of omega-3 fatty acids

- ALA or alpha-linolenic acid, is an 18-carbon chain and three cis double bonds. The first double bond is located in the n-3 position or at the omega end of the fatty acid.

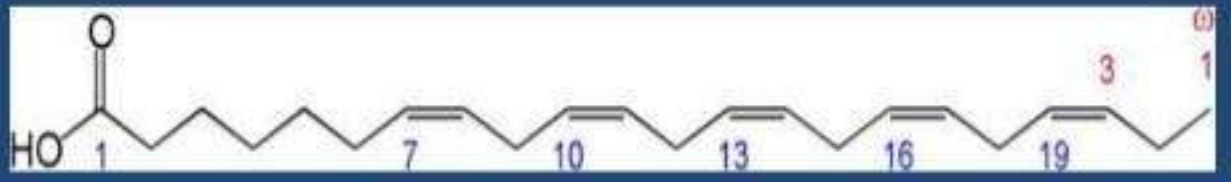
EPA or eicosapentaenoic acid contains a 20-carbon chain and five cis double bonds; the first double bond is located at the third carbon from the omega end. Therefore, EPA also is considered an omega-3 fatty acid.



Eicosapentaenoic Acid (EPA) (20:5 n-3)

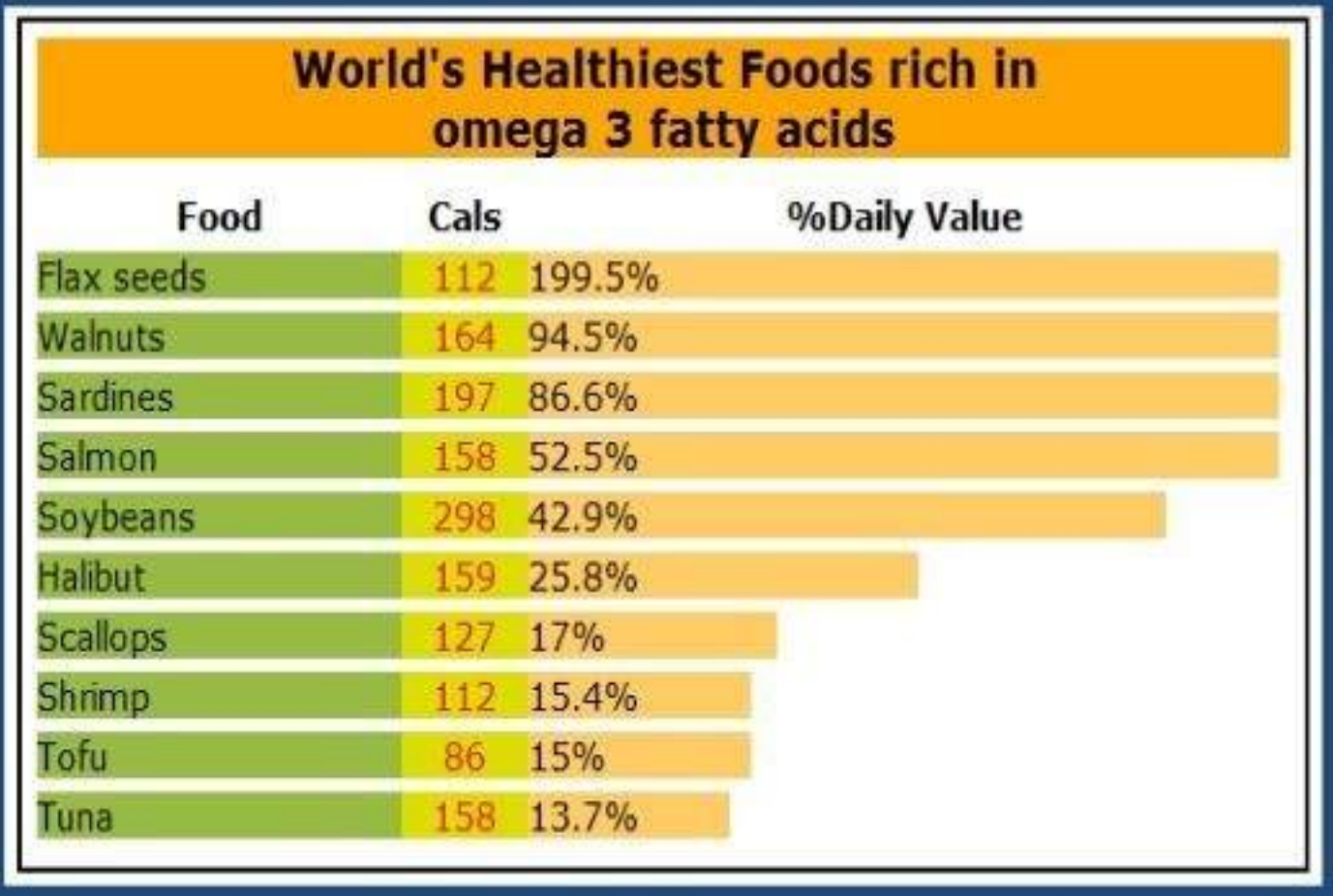
Figure by RMB

- DHA or docosahexaenoic acid is a 22-carbon chain with six cis double bonds; the first double bond is located at the third carbon from the omega end of the fatty acid.



Sources of these omega-3 fatty acids

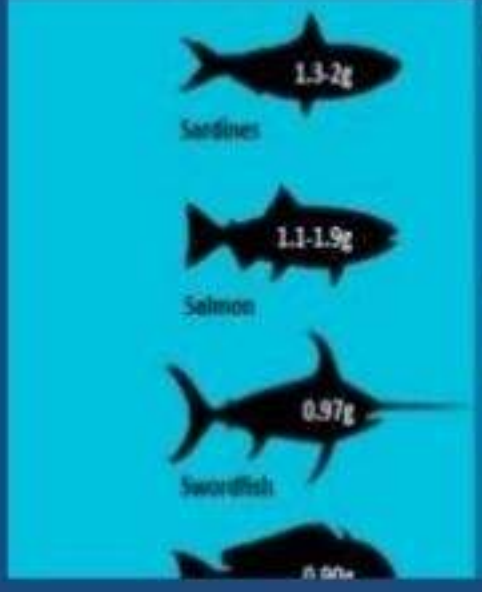
- ALA – Canola, Soybeans, Walnuts, and Flaxseed
- EPA – Oily fishes such as Cod Liver, Herring, Mackerel, Salmon, and Sardines
- DHA – Oily fishes such as Cod Liver, Herring, Mackerel, Salmon, and Sardines, and also are produced from algal fermentation


World's Healthiest Foods rich in omega 3 fatty acids
Food
Cals
%Daily Value
Flax seeds 112 199.5%
Walnuts 164 94.5%
Sardines 197 86.6%
Salmon 158 52.5%
Soybeans 298 42.9%
Halibut 159 25.8%
Scallops 127 17%
Shrimp 112 15.4%
Tofu 86 15%
Tuna 158 13.7%

Fish	Omega-3 fatty acids (mg/3oz)	Plant oils	Omega-3 fatty acids (g/tbsp)
Cod	134	Pumpkin seeds	0.051
Catfish	151	Olive oil	0.103
Haddock	203	Walnuts, black	0.156
Clams	241	Soybean oil	1.231
Shrimp	267	Canola oil	1.302
Flounder	426	Walnut oil	1.414
Pollock	460	Flaxseeds	2.350
Tuna, canned	733	Walnuts, English	2.574
Salmon	1825	Flaxseed oil	7.249

OMEGA-3

Good For:



Pregnancy Benefits:

- Neurological and early visual development
- Reduces risk of allergies
- Prevents pre-term labor and delivery
- Increases birth weight

Only 1 in 3 Americans include it in their daily food intake

CONCEIVE EASY

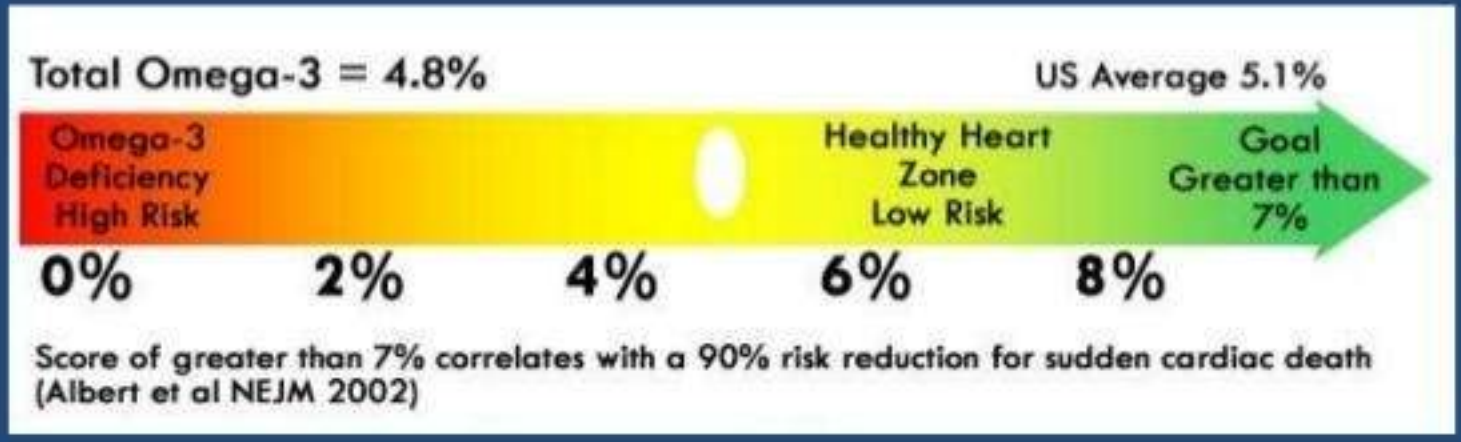

Poor source for Omega-3
Rich source for Omega-3
Red Snapper
(Ikan Merah)
200mg
Catfish
(Ikan Keli)
200mg
European Anchovies
(Ikan Bilis Eropah)
1400mg
Sardine (Sardin)
980-1700mg
Grouper
(Ikan Kertang)
300mg
Pacific Cod
(Kod Pasifik)
200mg
Indian Mackerel
(Kembong Fish)
340-1057mg
Atlantic Salmon
(Salmon Atlantik)
1400mg
The specific Omega-3 content is based on every 100g of the fish species shown above.[3]

Guideline portion amounts

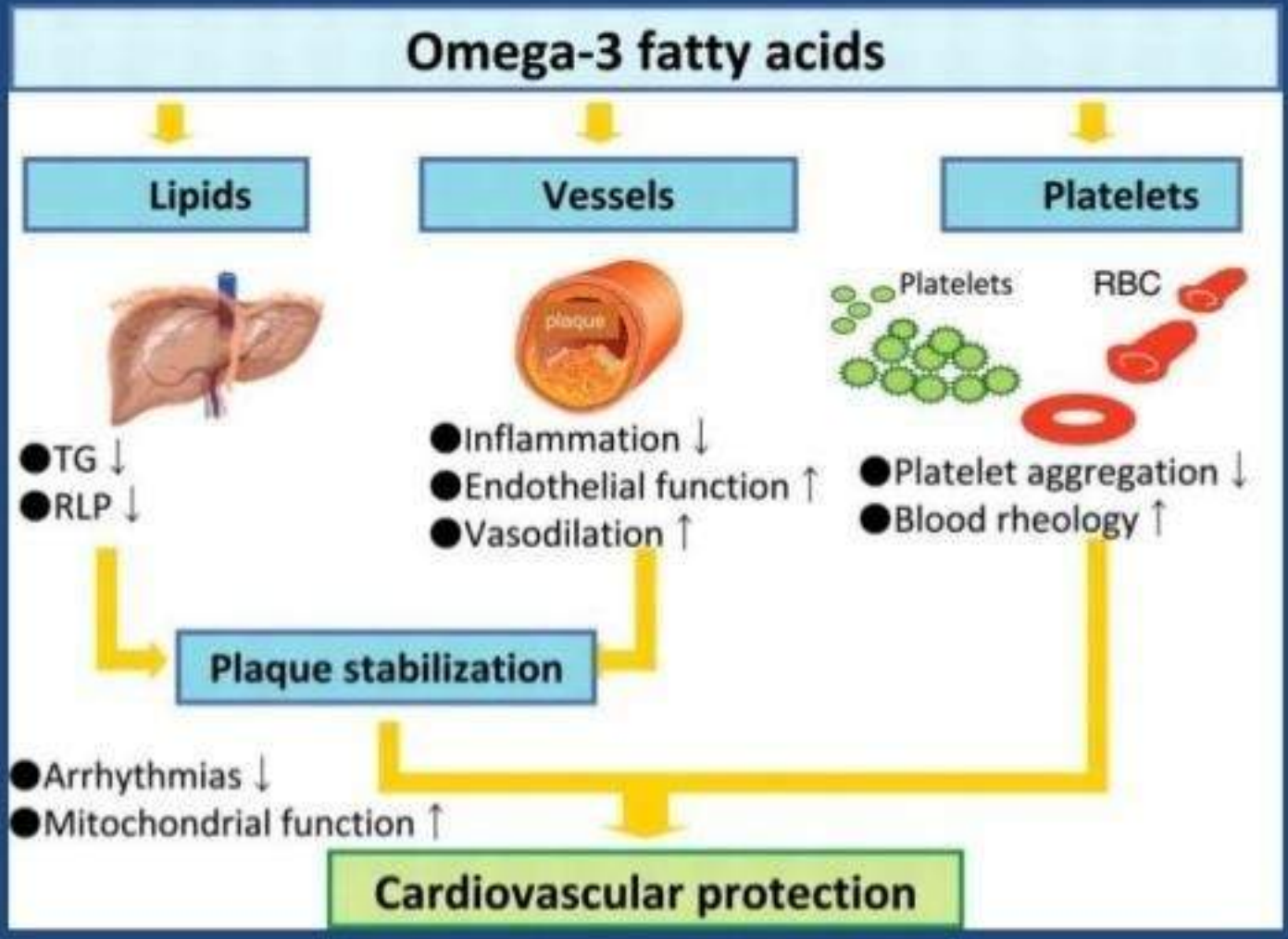
Age	One portion size
18 months to three years	¼ - ¾ small fillet or 1-3 tablespoons
four to six years	½ - 1 small fillet or 2 - 4 tablespoons
seven to eleven years	1 - 1 ½ small fillets or 3 - 5 tablespoons
12 years to adult	140g (5 oz) fresh fish or 1 small can oily.

Health benefits of omega-3s

- Omega-3 fatty acids correct imbalances in modern diets that lead to health problems. Eating foods rich in
- omega-3 fatty aids can help lower the risk of chronic diseases such as:


Total Omega-3 = 4.8%
US Average 5.1%
Omega-3 Deficiency High Risk
Healthy Heart Zone Low Risk
Goal Greater than 7%
0%
2%
4%
6%
8%
Score of greater than 7% correlates with a 90% risk reduction for sudden cardiac death (Albert et al NEJM 2002)

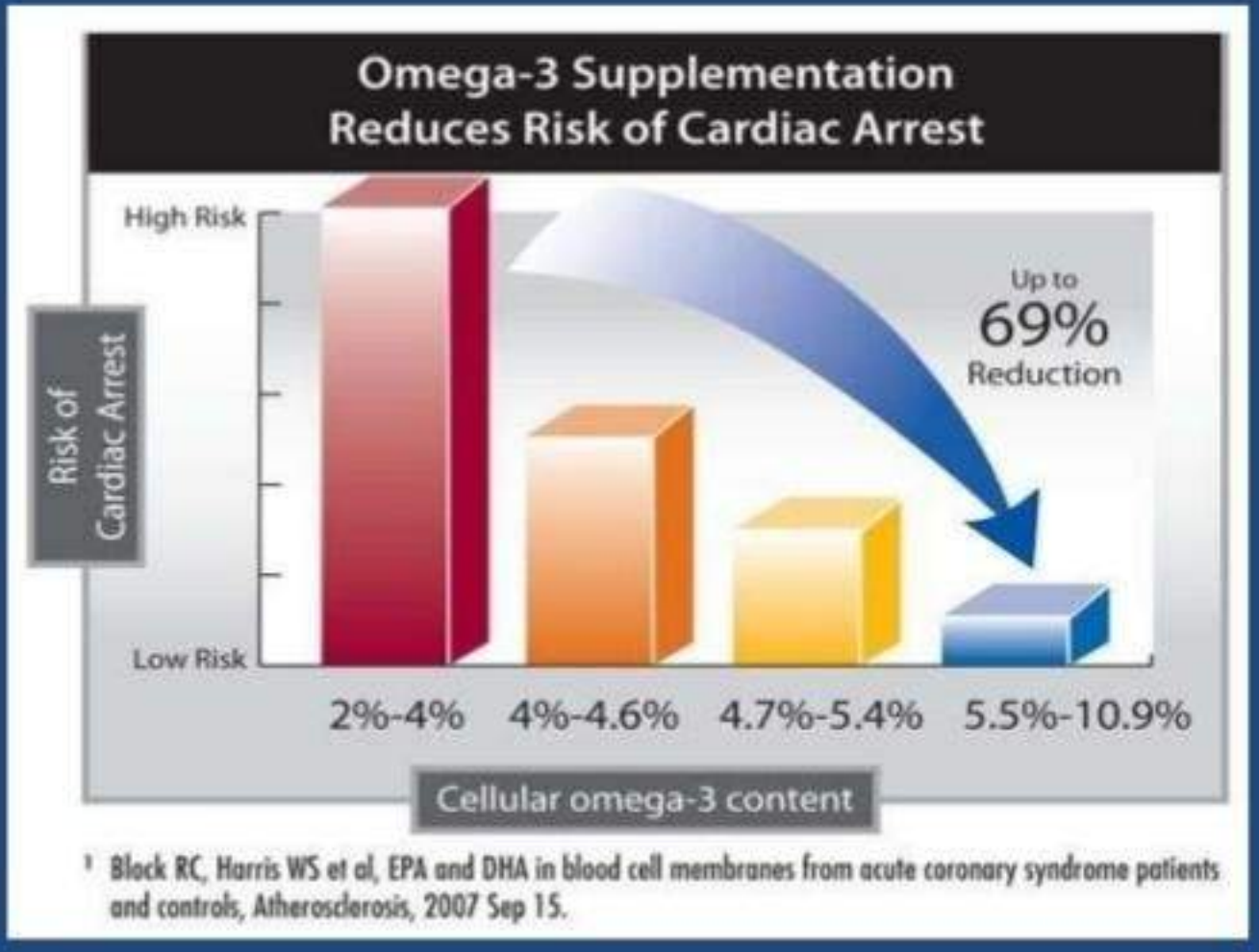

Health Benefits of Omega-3
Natural Hormone Replacement
Brain Function
Eyesight
Nervous System
Healthy Skin
Immune System
Joints
Heart
Reduces Inflammation Related to Many Diseases
Helps to Suppress Appetite
Liver (Cholesterol Levels)
Helps to Burn Stored Body Fat
Important for Pregnant Women & Nursing Mothers


Omega-3 fatty acids
Lipids
Vessels
Platelets
plaque
Platelets
RBC
●TG ↓
●RLP ↓
●Inflammation ↓
●Endothelial function ↑
●Vasodilation ↑
●Platelet aggregation ↓
●Blood rheology ↑
Plaque stabilization
●Arrhythmias ↓
●Mitochondrial function ↑
Cardiovascular protection

- Heart disease
- Stroke
- Cancer
- Lower LDL or “bad” cholesterol
- Studies show that diets high in amounts of EPA and DHA help with brain and eye development,
- Help to prevent Alzheimer’s disease.

- ALA – Studies show a diet high in ALA helps reduce heart disease and stroke by the following mechanisms:
- Reducing cholesterol and triglyceride levels
- Enhancing the elasticity of blood vessels
- Preventing the build-up of harmful fat deposits in the arteries.

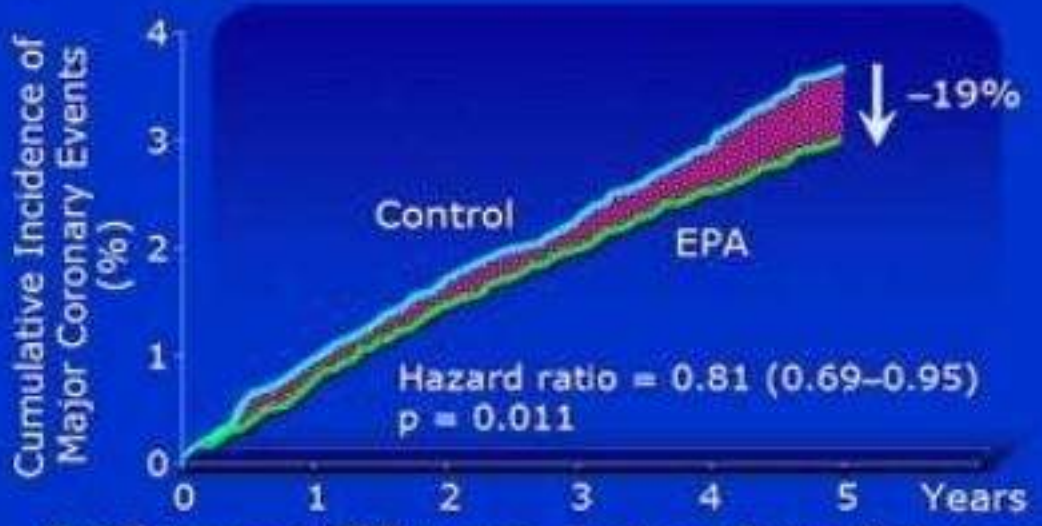
- A 10-year study correlated increased intakes of DHA/EPA as consumed by various population sectors with relative risk of heart related deaths.



[1] Block RC, Harris WS et al, EPA and DHA in blood cell membranes from acute coronary syndrome patients and controls, Atherosclerosis, 2007 Sep 15.

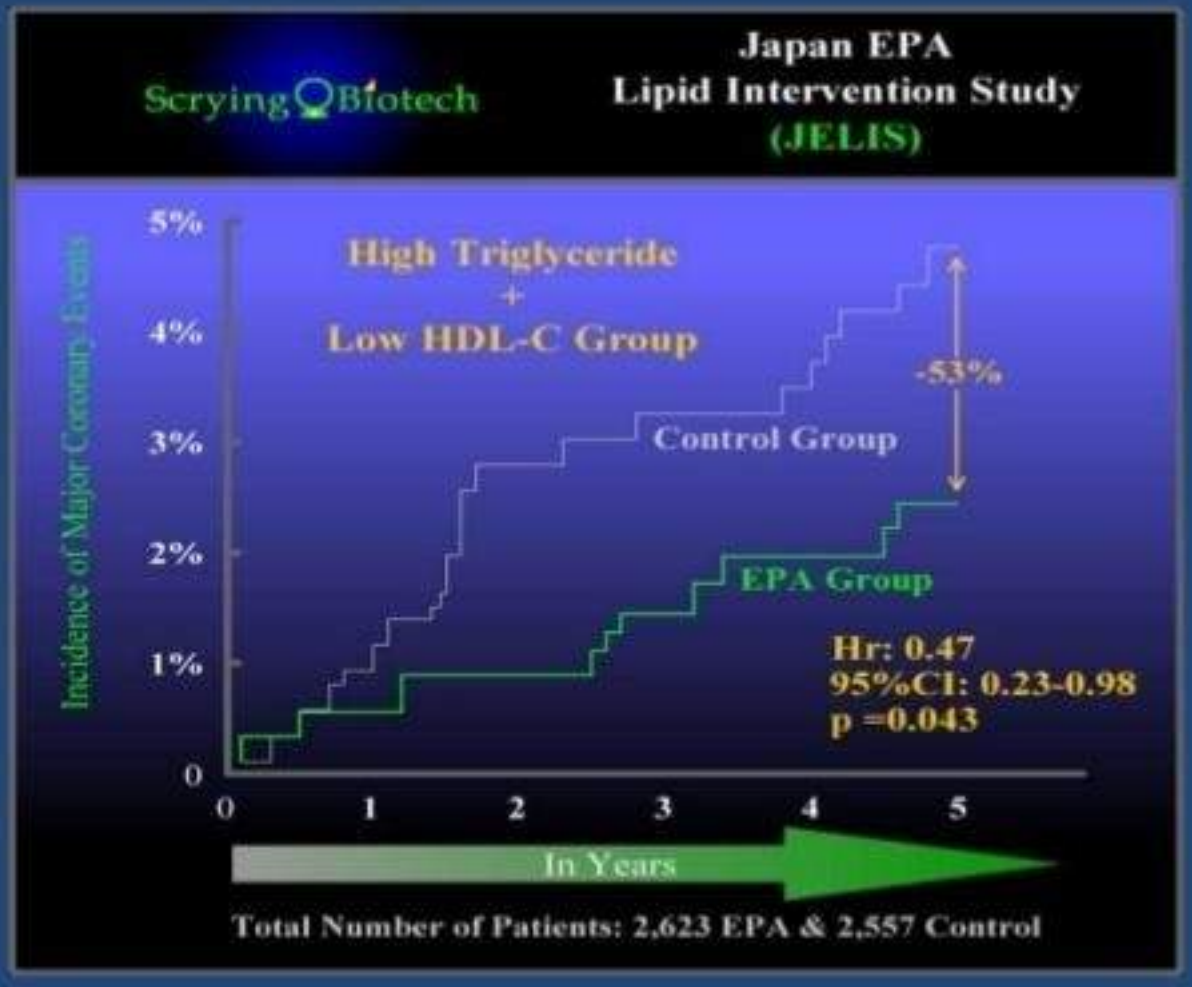
- Those who increased consumption of DHA/EPA up to 664 mg/day were associated with an approximate 40 percent reduction in cardiovascular disease and a significant reduction in all-cause mortality.3
- All infant formula is now supplemented with DHA.

Japan EPA Lipid Intervention Study (JELIS)



18,645 Japanese (70% women, mean age 61 years) randomized to statin alone or statin + EPA (1.8 g/d) and followed for 5 years

Yokoyama M. Presented at American Heart Association Scientific Sessions, Dallas, Texas, 14 November 2005.


Scrying Biotech
Japan EPA
Lipid Intervention Study
(JELIS)
Incidence of Major Coronary Events
5%
4%
3%
2%
1%
0
High Triglyceride
+
Low HDL-C Group
-53%
Control Group
EPA Group
Hr: 0.47
95%CI: 0.23-0.98
p =0.043
0
1
2
3
4
5
In Years
Total Number of Patients: 2,623 EPA & 2,557 Control

Omega-6 fatty acids

- Omega-6 fatty acid is also a polyunsaturated fat, essential for human health because it cannot be made in the body. For this reason, people must obtain omega-6 fatty acids by consuming foods such as meat,
- poultry, and eggs as well as nut and plant-based oils such as canola and sunflower oils.

Types of omega-6 fatty acids

- LA or linolenic acid is an unsaturated omega-6 fatty acid. Chemically, it is an 18-carbon chain. The first double bond is located at the sixth carbon from the omega end of the fatty acid.

- AA or Arachidonic acid is a 20-carbon chain. Its first double bond is located at the sixth carbon from the omega end of the fatty acid.

Sources of omega-6 fatty acids

- LA – Soybean oil, Corn oil, Safflower Oil, Sunflower Oil … Peanut Oil, Cottonseed oil, and Rice Bran Oil
- AA – Peanut Oil, Meat, Eggs, and Dairy Products

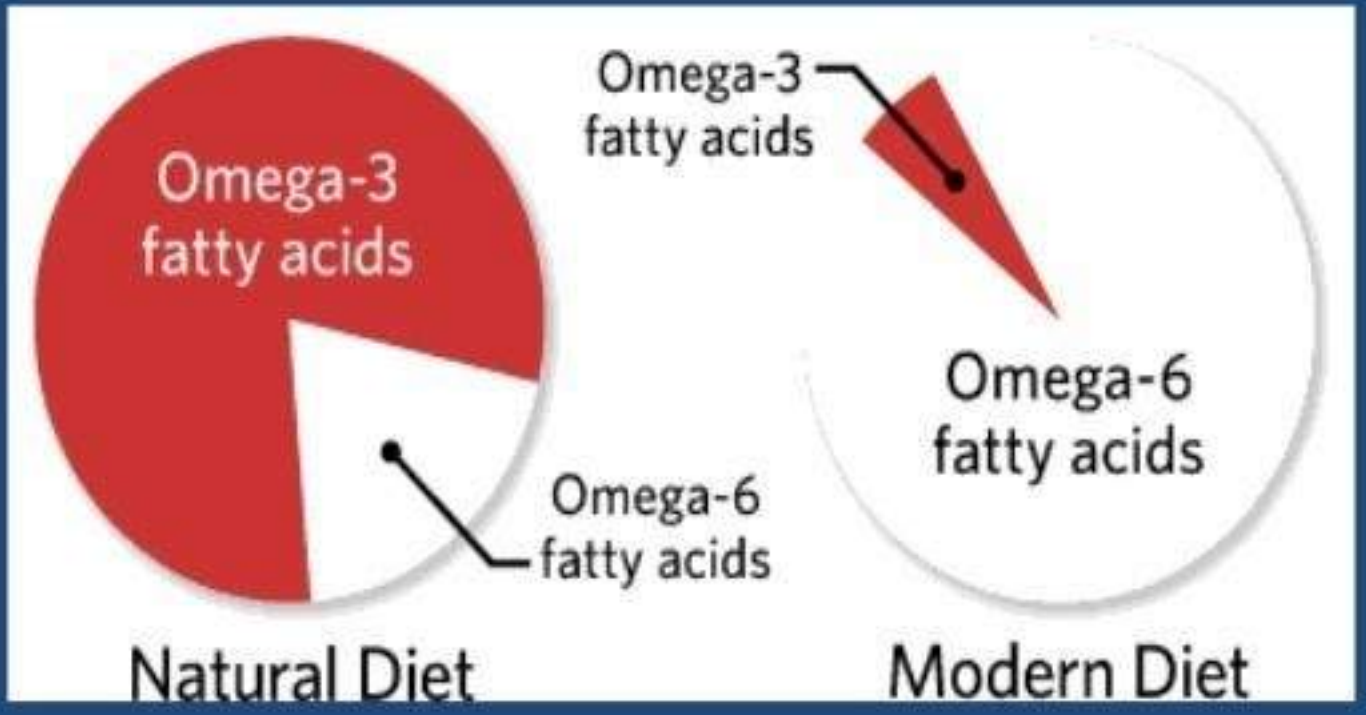
Benefits of omega-6 fatty acids

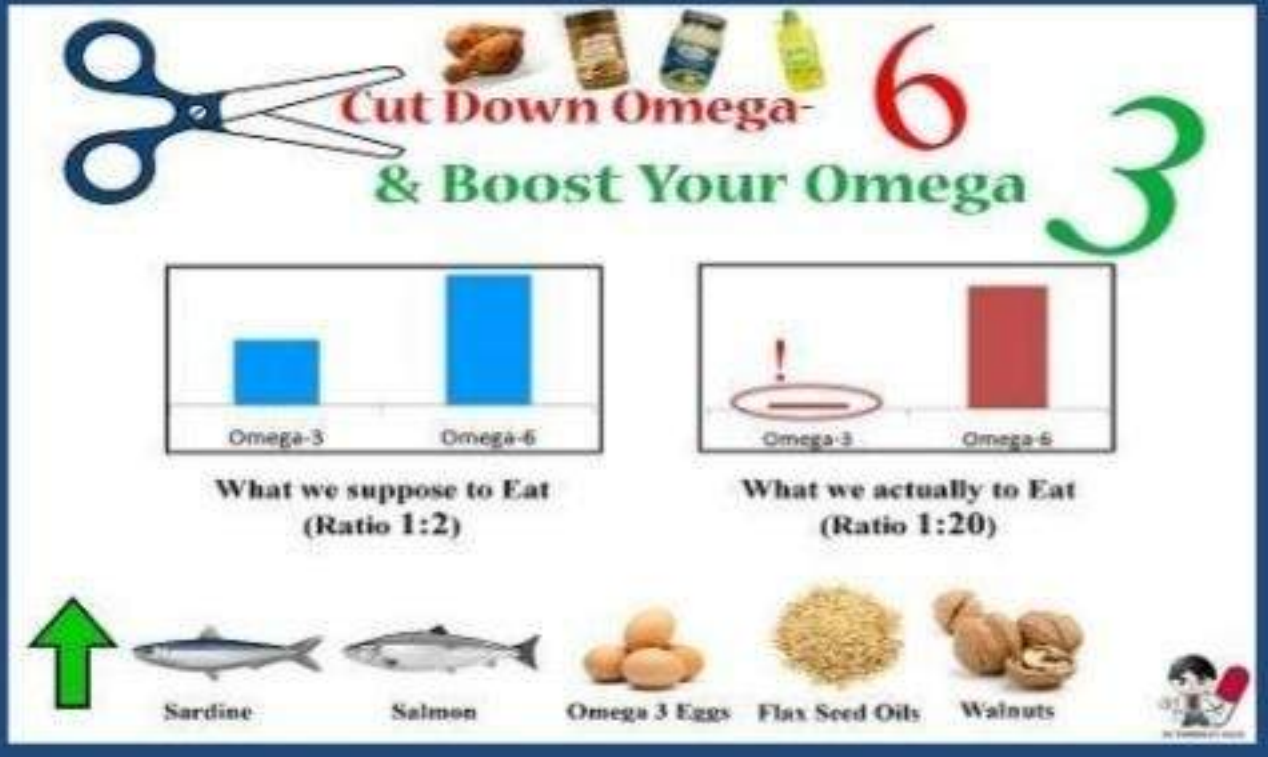
- Most omega-6 fatty acids are consumed in the diet from vegetable oils such as linoleic acid. Excessive amounts of linoleic acid can contribute to inflammation and result in heart disease, cancer, asthma, arthritis, and depression.

- Omega-3 and Omega-6 fatty acids: Striking the Balance
- By finding a balance between omega-3 and omega-6 fatty acids in the diet, both substances can work together to promote health.

- A healthy diet should consist of roughly two to four times more omega-6 fatty acids than omega-3 fatty acids.

- A typical North American diet may contain 11 to 30 times more omega-6 fatty acids than omega-3 fatty acids, contributing to the rising rate of inflammatory disorders in the United States.


Omega-3
fatty acids
Omega-6
fatty acids
Natural Diet
Omega-3
fatty acids
Omega-6
fatty acids
Modern Diet


Cut Down Omega-6
& Boost Your Omega 3
Omega-3
Omega-6
What we suppose to Eat
(Ratio 1:2)
!
Omega-3
Omega-6
What we actually to Eat
(Ratio 1:20)
Sardine
Salmon
Omega 3 Eggs
Flax Seed Oils
Walnuts

Omega-9 fatty acids

- Omega-9 fatty acids are from a family of unsaturated fats that commonly are found in vegetable and animal fats.

- This monounsaturated fat is described as omega-9 because the double bond is in the ninth position from the omega end. These fatty acids are also known as oleic acids can often be found in canola, sunflower, olive, and nut oils.

- Unlike omega-3 and omega-6 fatty acids, omega-9 fatty acids are produced by the body, but are also beneficial when they are obtained in food.

Types of omega-9 fatty acids

- Oleic acid is a main component of canola oil, sunflower oil, olive oil, and other monounsaturated fats, many of which are used as a solution for reducing bad fats in cooking oils.

Sources of omega-9 fatty acids

- Oleic acid
- Canola oil
- Sunflower Oil
- Almonds oils for

Health benefits of omega-9 fatty acids

- Research has shown that omega-9 fatty acids, commonly referred to as monounsaturated fatty acids, can help reduce the risk of cardiovascular disease and stroke.

Mechanism of action

- Because omega-9 fatty acids have been shown to increase HDL ("good") cholesterol and decrease LDL ("bad") cholesterol, they help eliminate plaque buildup in the arteries, which causes heart attack and stroke.

Safety of essential fatty acids

- Essential fatty acids are extremely safe especially if taken from plant sources

- There has been publicity about chemicals that may be harmful in some types of fish.
- Shark, swordfish and marlin may contain concentrated sources of mercury that may be harmful to the developing baby's nervous system and so should be avoided by women who are pregnant or planning a baby, and by all children under 16 years.


WARNING
CONTAMINATED FISH
FISH IN THESE WATERS ARE
CONTAMINATED WITH
DANGEROUS LEVELS OF
POISONOUS MERCURY
DO NOT EAT FISH
CAUGHT IN THESE WATERS

All other adults, including breastfeeding women, should eat no more than one portion of these particular fish a week.

- Women past childbearing age or not intending to have children, men and boys can eat up to four portions of all other oily fish a week.

- Women who are pregnant or breastfeeding, or likely to become pregnant, and girls who may become pregnant in the future, can safely have up to two portions of oily fish a week.

What if I don't like/eat fish?

- People who do not eat fish can get omega-3 from the following foods: nuts and seeds e.g. walnuts and pumpkin seeds; vegetable oils e.g. rapeseed and linseed; soya and soya products e.g. beans, milk and tofu; and green leafy vegetables.

- Omega-3 enriched foods Some foods have omega-3 added to them and can be useful sources, especially for vegetarians and

- others who avoid fish. However these tend to be small amounts but do include eggs and some brands of milk, yoghurt, bread and spreads. These foods may help to increase omega-3 intake.
- Supplements It is best to try and get omega-3 from foods but the following advice may be helpful if you find this difficult and wish to take a supplement:

Look for omega-3 oil rather than fish liver oil

- Check the vitamin A content -The Scientific Advisory Committee on Nutrition (SACN) advises that if you take supplements containing vitamin A, you should not have more than a total of 1.5mg (1500ug) a day from food and supplements combined


Kintek
Natural Orange Flavor
Omega-3
Fish Oil Concentrate 1000mg
May promote Cardiovascular Health and Brain Function.
180 SOFTGELS